Egypt-Japan University of Science and Technology
الجامعة المصرية اليابانية للعلوم و التكنولوجيا
エジプト日本科学技術大学

EECE
SCHOOL OF ENERGY RESOURCES, ENVIRONMENT, CHEMICAL AND PETROCHEMICAL ENGINEERING

Intro. Energy, Environmental and Chemical Eng. (CPE 211)

Assignment#3 "Environmental Engineering"

Question#2 (11 marks)

This Table lists the chemical component of a solid waste sample, with a total mass= 120 kg

Component	Percentage of total mass (%)	Moisture content (%)	Percentage of dry mass (%)					
			C (12)	H (1)	O (16)	N (14)	S (32)	Ash
Food waste	(a)	60	46.8	7.2	37.8	3.4	0.3	4.5
Paper	30	75	48.0	6.4	37.6	(b)	0.4	5.0
Card board	10	20	49.5	6.5	42.2	0.2	0.1	1.5
Plastics	40	6	42.0	6.0	45.0	0.5	0.5	6.0

Dulong's formula for energy content estimation (in kJ/kg): E=337C+1428(H-O/8)+9S

1.	The value of (a)= …							
	(A)	10	(B)	20	(C)	30	(D)	40
2.	The value of (b)= …							
	(A)	3.2	(B)	2.6	(C)	1.7	(D)	3.3
3.	Assuming the chemical composition has S = 1; the number of moles of C= …							
	(A)	134	(B)	374	(C)	289	(D)	43
4.	Assuming the chemical composition has S = 1; the number of moles of H= …							
	(A)	1044	(B)	323	(C)	2045	(D)	765
5.	Assuming the chemical composition has S = 1; the number of moles of O= …							
	(A)	594	(B)	233	(C)	347	(D)	486
6.	Assuming the chemical composition has S = 1; the number of moles of N= …							
	(A)	2.4	(B)	6.1	(C)	8.3	(D)	3.3
7.	Assuming the chemical composition has N = 1; the number of moles of C= …							
	(A)	47	(B)	31	(C)	78	(D)	13
8.	Assuming the chemical composition has N = 1; the number of moles of H= …							
	(A)	170	(B)	138	(C)	105	(D)	231
9.	Assuming the chemical composition has N = 1; the number of moles of O= …							
	(A)	39	(B)	79	(C)	92	(D)	630
10.	Assuming the chemical composition has N = 1; the number of moles of S= …							
	(A)	0.2	(B)	1.5	(C)	2.4	(D)	3.7
11.	The energy content in the solid waste= … kJ/kg							
	(A)	11928	(B)	3283	(C)	15433	(D)	8654

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